



FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 01.08.2026

In addition to part I (the General Handout for all courses, appended to the Time table), this section provides further specific details regarding the course.

Course No : SCM G514
Course Title : Logistics and Procurement
Instructor-in-Charge : AMIT KUMAR JAIN
Instructor(s) : NA
Tutorial/Practical Instructors: NA

1. Course Description:

Supply chain network design; Supply chain Relationships; Freight modes; Logistics planning; Shipment execution; Transportation sourcing and procurement; Freight Billing and Payments; Omni-Channel Distribution; Global Trade Management (GTM); Warehousing; RFID in supply chain; Warehouse Picking; Yard Management; Strategic Sourcing; Agreements, Supplier Management; Supplier Risk Management, Spend Analysis and Optimization; AI and use cases in procurement; Inventory management-Concepts and classifications; Types of Inventory Transactions; Managing Uncertainty; cost management; inventory control techniques; Product genealogy traceability and recall management; Replenishment; Material movements across facilities; CPFR concept; Project Based Supply Chain; Digital Technologies in SCM; Integration with other SCM systems.

2. Scope and Objective of the Course:

This course will give students a broad overview of the fundamental theories regarding various logistics functions and their overall management. With this course, students will be able to approach logistics management with a holistic view and understand, analyse, and coordinate various functions within a coherent supply chain framework.

- This will help students understand the various components of procurement – sourcing, contracting, supplier management, qualification management, purchasing of direct and indirect materials, spend analysis, and its integration with other management systems/applications.
- This will help explore the importance of inventory management, the key components involved, and the best practices that businesses can implement to improve their inventory management processes.

3. Text Books:

T1: Vinod V. Sople; Logistics Management, 3rd ed.; Pearson Education, 2012, ISBN: 9788131768624.

T2: Donald J. Bowersox, David J. Closs, M. Bixby Cooper, John C. Bowersox; Supply Chain Logistics Management, 5th ed.; McGraw-Hill Education (India), 2024, ISBN: 9789355327215.

T3: Roberta S. Russell, Bernard W. Taylor, Venkataramanaiah Saddikuti, Pavan Kumar Gudavalleti; Operations and Supply Chain Management, 10ed (An Indian Adaptation); Wiley, 2023, ISBN: 9789354644092.

T4: Satish C. Ailawadi, P. Rakesh Singh; Logistics and Supply Chain Management, 3rd ed.; PHI Learning Pvt. Ltd., 2021, ISBN: 9788194800286.



4. Reference Books:

R1: Martin Christopher; Logistics and Supply Chain Management, 6th ed.; FT Publishing International (Pearson), 2023, ISBN: 9781292416205.

R2: Hamdy A. Taha; Operations Research: An Introduction, 11th ed.; Pearson, 2022, ISBN: 9780137625727.

R3: Arjan J. van Weele, Frank Rozemeijer; Procurement and Supply Chain Management, 8th ed.; Cengage Learning, 2022, ISBN: 9781473779112.

R4: Ronald H. Ballou, Samir K. Srivastava; Business Logistics/Supply Chain Management, 5ed; Pearson Education, 2007, ISBN: 9788131705841.

R5: Carlos Mena, Martin Christopher, Remko van Hoek; Leading Procurement Strategy: Driving Value Through the Supply Chain; Kogan Page, 2014, ISBN: 9780749470340.

R6: Selected journal papers shared during lectures.

5. Course Plan:

Week No. Lecture Session	Module	Topics	Ref.	Learning outcomes
1 L: 1-3	Foundations of Logistics	Course introduction. Introduction to logistics: definition, scope, objectives, components, and importance of logistics. Types of logistics (procurement, production, distribution, reverse, after-sales, global, domestic). Role of logistics in manufacturing and the economy. Difference between logistics and supply chain management. Supply chain relationships. Case study.	T1, Ch 1 Lecture Notes	Understand the fundamentals of logistics, its types, key components, supply chain relationships, and its role in manufacturing and business operations.
2 L: 4-6	Transportation Network Modelling	Mathematical formulation of transportation and distribution models. Quantitative techniques to obtain an initial basic feasible solution for balanced and unbalanced transportation problems. Northwest Corner Method (NWCM). Least Cost Method. Hands-on numerical examples.	R2, Ch 5 Lecture Notes	Formulate and solve balanced and unbalanced transportation problems using basic quantitative methods.
3 L: 7-9	Transportation Network Optimisation	Quantitative Techniques to determine optimal movement of goods between plants, warehouses, and customers. Vogel's Approximation Method (VAM) for obtaining high-quality initial solutions. Comparison of	R2, Ch 5 Lecture Notes	Apply quantitative techniques to determine optimal transportation plans and interpret the results to support



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		different initial solution methods and their efficiencies. Optimal transportation plans. MODI method. Degeneracy. Solving maximisation transportation problems. Spreadsheet implementation. Interpreting results for managerial decision-making.		effective logistics decision-making.
4 L: 10-12	Routing and Distribution Optimisation	Travelling Salesman Problem (TSP). Vehicle Routing Problems (VRP). Distribution routing strategies. Logistics routing applications. Case studies and optimisation exercises.	R2, Ch 11 Lecture Notes	Apply routing and optimisation techniques to solve logistics distribution and vehicle routing problems.
5 L: 13-15	Integrated Transportation Decision Making	Transportation network planning. Integration of transportation with distribution planning. Carrier selection. Transportation performance measures. Strategic transportation decisions and industry applications.	T1, Ch 8 Lecture Notes	Apply transportation planning, carrier selection, and performance measures to support effective distribution and strategic logistics decisions.
6 L: 16-18	Inventory Management Fundamentals	Introduction to inventory management and its role in supply chains. Objectives and functions of inventory. Types and classifications of inventory (raw materials, WIP, finished goods, MRO). Inventory transactions, including receipts, issues, transfers, returns, adjustments, and stock verification. Inventory cost components include ordering, holding, shortage, and purchasing costs. Hands-on exercises on inventory classification and inventory transaction scenarios with practical numerical examples.	T3, Ch 13 T2, Ch 7 Lecture Notes	Understand inventory concepts, classifications, transactions, and costs, and apply them to practical inventory management problems.
7 L: 19-21	Deterministic Inventory Models	Continuous demand, instantaneous replenishment model. Inventory replenishment concepts and assumptions of the EOQ model. Development and mathematical derivation of the EOQ model. Analysis of total inventory cost, reorder point, and inventory cycle. Sensitivity analysis and managerial interpretation of EOQ. Step-by-step solution of a variety of numerical problems and hands-on implementation.	T3, Ch 13 T2, Ch 7 Lecture Notes	Apply the EOQ model to determine optimal inventory replenishment decisions and evaluate their cost implications.
8	Advanced Inventory Models	Economic Production Quantity (EPQ) model and its applications. Quantity	T3, Ch 13	Apply EPQ and quantity discount models to determine optimal



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L: 22-24		Discount models. Selection of the optimal purchasing quantity considering price breaks and total cost. Comparison of EOQ and EPQ models. Comprehensive numerical examples, case studies, and hands-on.	T2, Ch 7 Lecture Notes	production and purchasing quantities for effective inventory decisions.
9 L: 25-27	Inventory Control under Uncertainty	Demand and lead-time uncertainty. Reorder Point (ROP), Safety Stock, and Service Level concepts. Inventory replenishment policies under uncertain demand. Shortage and Backordering models, including planned shortages and cost trade-offs. Extensive numerical problem-solving, case-based exercises, and hands-on analysis of inventory decisions under uncertainty. Inventory control techniques. Cycle counting and inventory performance measures	T3, Ch 13 T2, Ch 7 Lecture Notes	Apply inventory control techniques to effectively manage demand uncertainty, safety stock, shortages, and inventory performance.
10 L: 28-30	Logistics Network Design	Supply chain network design. Facility location decisions in logistics. Location problems. Location-allocation problems. Logistics planning. Material movement across facilities. Facility layout planning. Quantitative models for facility layout.	T2, Ch 11 Lecture Notes	Apply quantitative techniques for supply chain network design, facility location, layout, and material movement decisions.
11 L: 31-33	Transportation Systems and Distribution	Freight transportation modes (road, pipeline, rail, air, sea, and multi-model transport). Factors influencing transportation. Shipment execution. Transportation documentation. Concept of containerisation. Transportation sourcing and procurement. Freight billing and payment. Global Trade Management (GTM). Omni-channel distribution.	T2, Ch 3 Lecture Notes	Understand freight transportation modes, shipment execution, documentation, sourcing, billing, global trade, and omni-channel distribution.
12 L: 34-36	Warehouse and Distribution Operations	Warehousing: challenge, functions, options, site selection, layout design, decision model, costing, strategies, performance parameters. Yard Management. RFID applications. Product genealogy, traceability, and recall management.	T1, Ch 3 Lecture Notes	Understand warehousing decisions, technologies, and performance measures for efficient storage, yard management, and product traceability.
13 L: 37-39	Procurement and Supplier Management	Strategic sourcing. Supplier agreements. Supplier relationship management. Supplier risk management. Spend analysis and	R3, Ch 1 and 2 Lecture Notes	Apply strategic sourcing, supplier management, spend analysis, and AI tools to improve procurement



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		optimisation. AI applications in procurement.		decisions and manage supplier risks.
14 L: 40-42	Digital Supply Chains and Enterprise Integration	Collaborative Planning, Forecasting, and Replenishment (CPFR). Digital technologies in SCM. Integration with ERP and other SCM systems. Project-based supply chains. Emerging trends: AI for forecasting and related areas. Integrated logistics case studies.	R1, Ch 16 Lecture Notes	Understand collaborative planning, digital technologies, systems integration, and emerging trends to support effective supply chain management.

6. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time*	Nature of component Close Book (CB) / Open Book (OB)
Mid-Semester Test	90 min	25	08/10 AN2	CB
Quiz	30 min	10	To be announced in class	CB, Best 2 of 3
Project	-	30	To be announced in class	OB
Comprehensive Examination	3 h	35	12/12 AN	CB + OB
Total		100		

*Date and time are tentative and subject to change. Stay updated with the information from the exam office.

7. Chamber Consultation Hour: Chamber no 2219-A; Wednesday, 5.00 PM to 6.00 PM

8. Notices: All the notices regarding the course will be emailed or displayed on the Nalanda or will be displayed on the Mechanical Engineering Department Notice Board only.

9. Make-up Policy: Make-up will be permitted only for genuine cases with prior permission. For the Quiz and Project, no make-ups will be granted.

10. Attendance Policy: Each student is expected to attend all classes and participate in all evaluation components.

11. Minimum passing Marks criteria: Failure to appear for the mid-semester/comprehensive exam. Scoring less than 20 marks out of 100 or a suitable modification at the discretion of the IC, with the concurrence of the Department.

12. Note (if any): NA

Instructor-in-charge

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